Foundation Models in Single-Cell Biology

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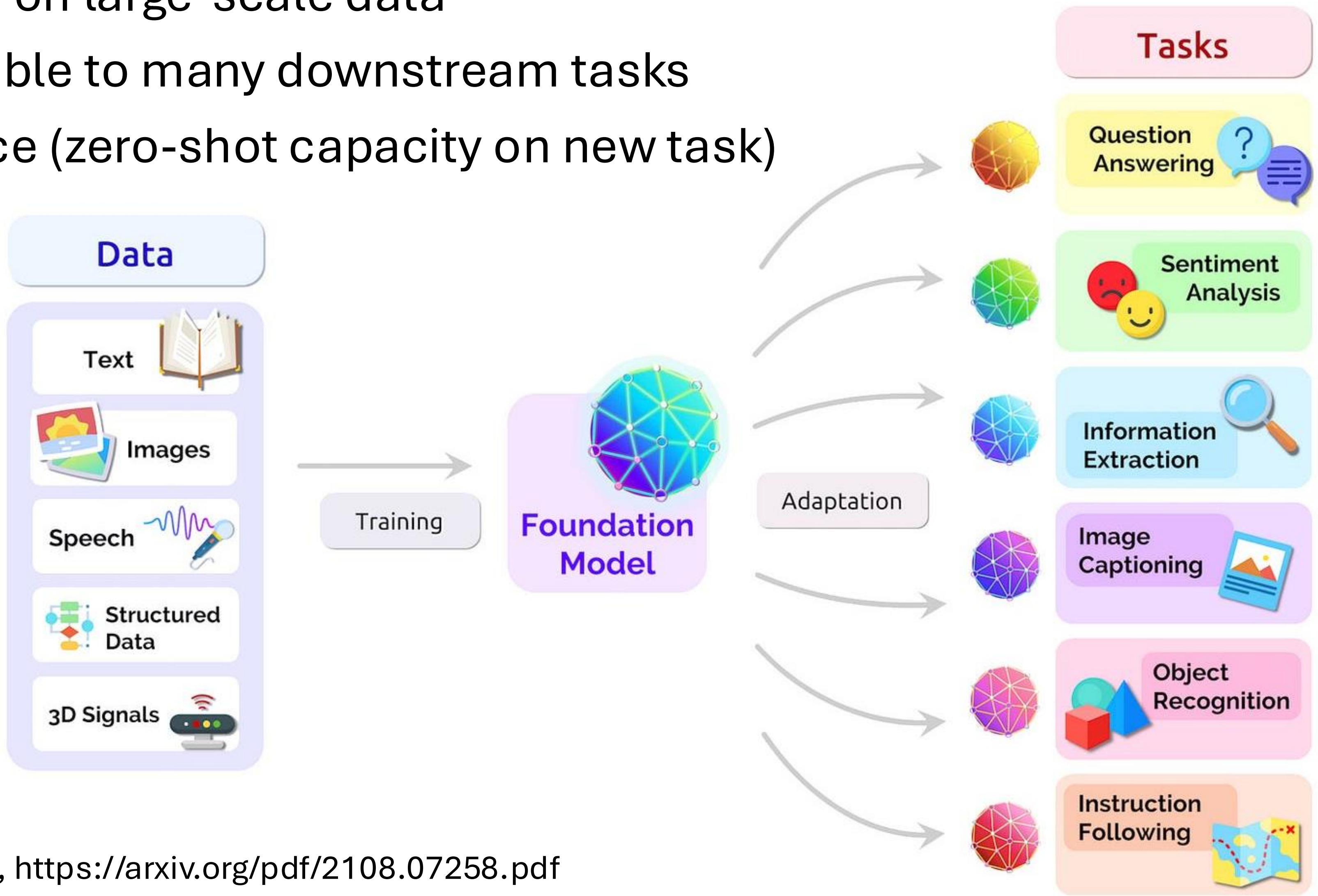
Acknowledgments

This file is for the educational purpose only. Some materials (including pictures and text) were AI-generated or taken from the Internet at the public domain.

What is a foundation model

- ➢Large model trained on large-scale data
- ➢Promptable, adaptable to many downstream tasks
- ➢Emerging intelligence (zero-shot capacity on new task)

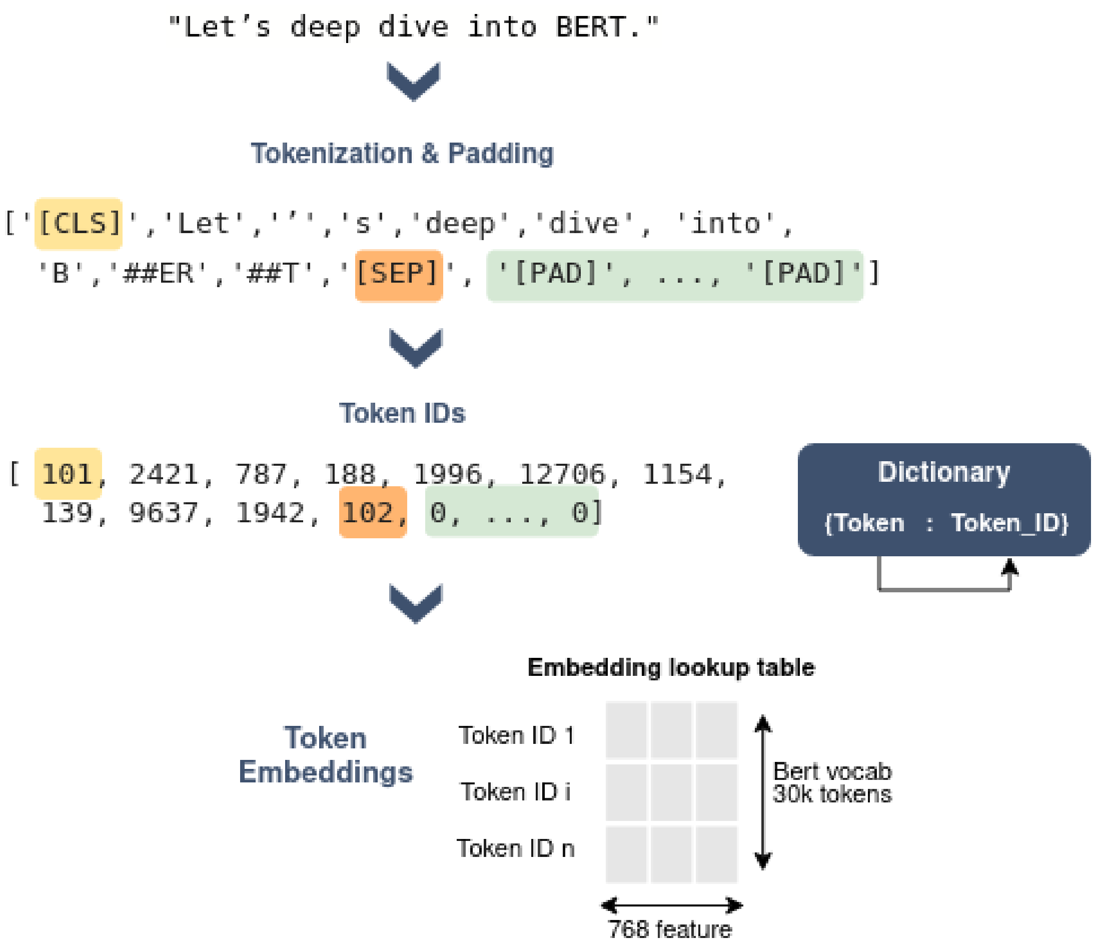


Bommasani et al, 2022, https://arxiv.org/pdf/2108.07258.pdf

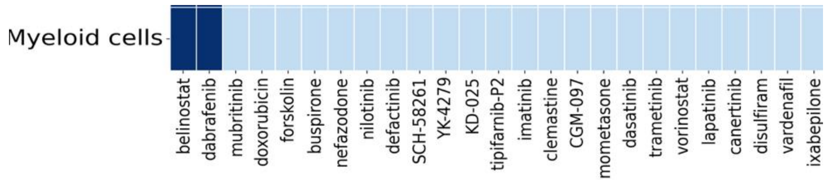
LLM vs. single-cell foundation model

Large Language Model (LLM)	Single-cell Foundation Model (scFM)
Tokens are words/subwords	Tokens may be genes, binned expression, cells, patches, or spatial neighborhoods
Word order is meaningful	Gene order is artificial or induced by rank/expression/genomic position
Text has grammar and context	Cells have gene programs, pathways, perturbation responses, batch effects
Pretraining data are huge and natural	Single-cell atlases are large but more redundant and biased by tissue, platform, disease, lab, species
Decoder-only, hard prompting (instruction)	Encoder-only, embedding and soft prompting (fine-tuning with adaptors)

From LLM to scLLM

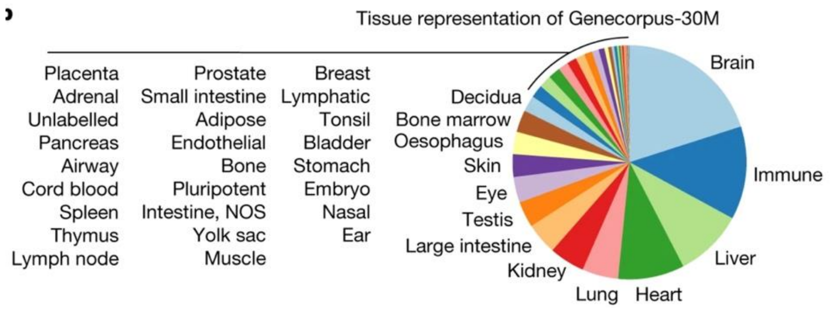


Training: auto-regressive vs. masking



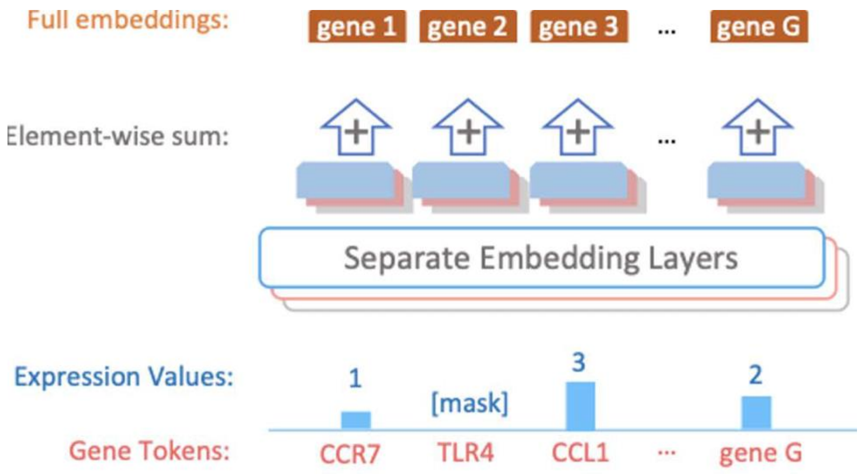
Expression profile of a single cell

Tokenization



Gene dictionary/ vocabulary

Gene IDs



Scales of Single-cell LLMs

Model	***Paper***	***# Parameters***	***Architecture***	***Training Data***	***Downstream Tasks***
scGPT	Cui et al. bioRxiv 2023	51 million	autoregressive transformer	33 million normal human cells (51 tissues, 441 studies)	cell-type annotation, multi-batch integration, multi-omic integration, perturbation prediction, and GRN inference
scBERT	Yang et al. Nature MI 2022	5 million	Performer (allowing for longer inputs)	209 human single-cell datasets comprising 74 tissues with 1M+ cells	cell type annotation
Geneformer	Theodoris et al. Nature 2023	40 million	Transformer	Genecorpus-30M-29.9 million human single-cell transcriptomes	gene dosage sensitivity, chromatin, network dynamics,
scFoundation	Hao et al. bioRxiv 2023	100 million	Transformer (w/ trick to reduce # words)	50 million human cells (100+ tissue types, normal and disease)	clustering, perturbation prediction, drug response

Slide credit: Jian Ma (www.cs.cmu.edu/~jianma/talks/LLM-CGSI-slides-public.pdf)

Five families of single-cell foundation models

Gene/expression token models

scBERT, Geneformer, scGPT, scFoundation, CellFM

Cell as a sequence or set of gene-expression tokens

Cell embedding / atlas query

CellPLM, SCimilarity, UCE, scTab

Reusable cell representations for mapping and retrieval

Knowledge-informed / cross-species

GeneCompass, SATURN-like strategies

Use priors and orthology to transfer biological structure

Spatial and multimodal models

Nicheformer, Novae, OmiCLIP/Loki, STPath

Model tissue context, morphology and spatial neighborhoods

Perturbation-response models

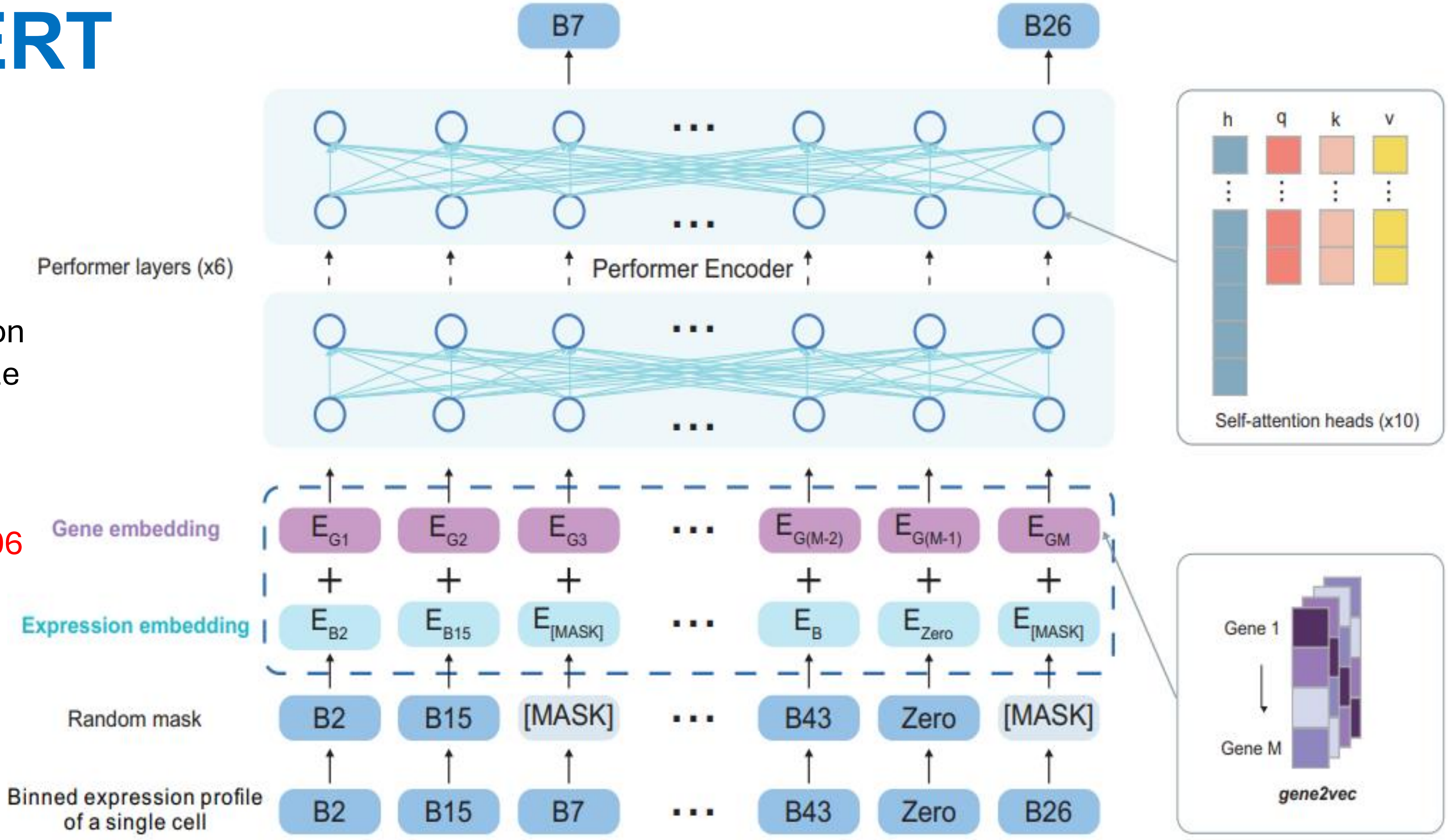
GEARS hybrids, Stack, State

Predict response to gene, drug or environmental changes (virtual cell)

scBERT

Binning expression profile to vectorize initial expression embedding

Vocabulary: 16906

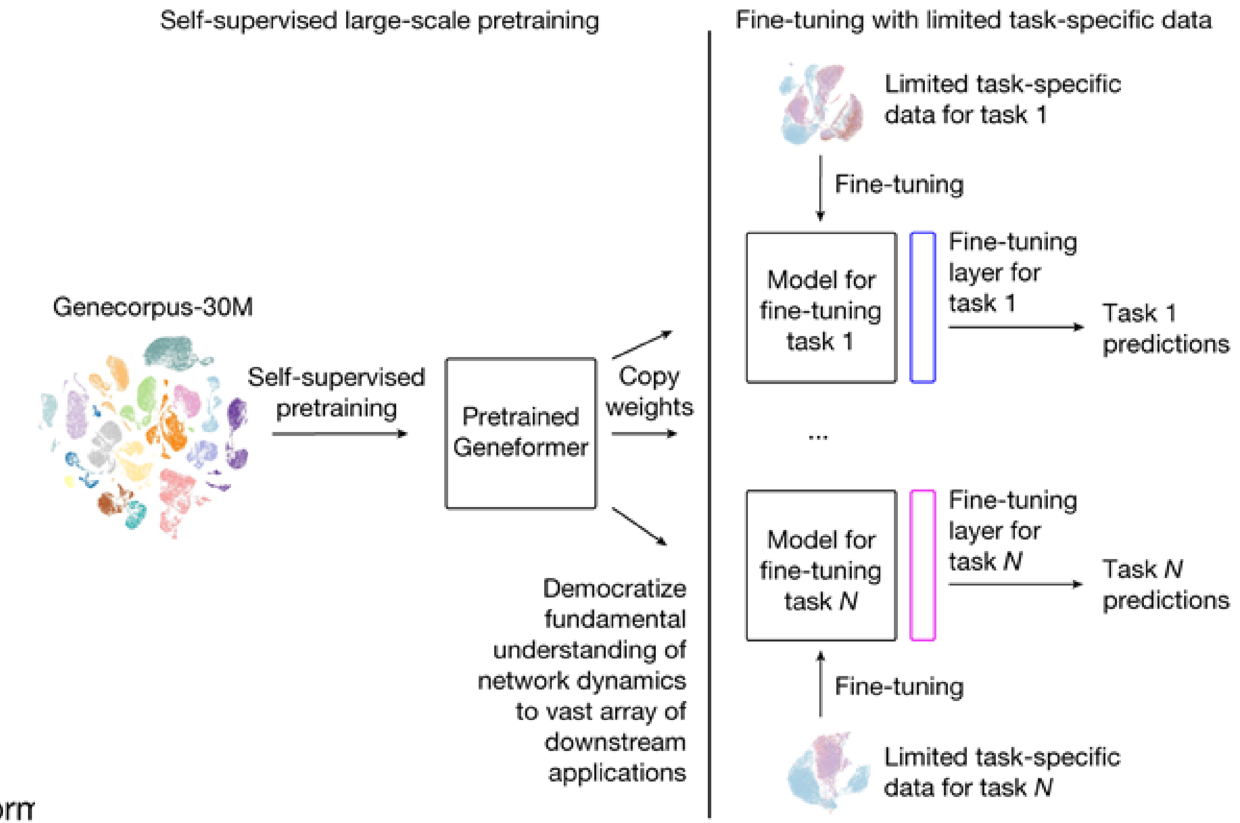


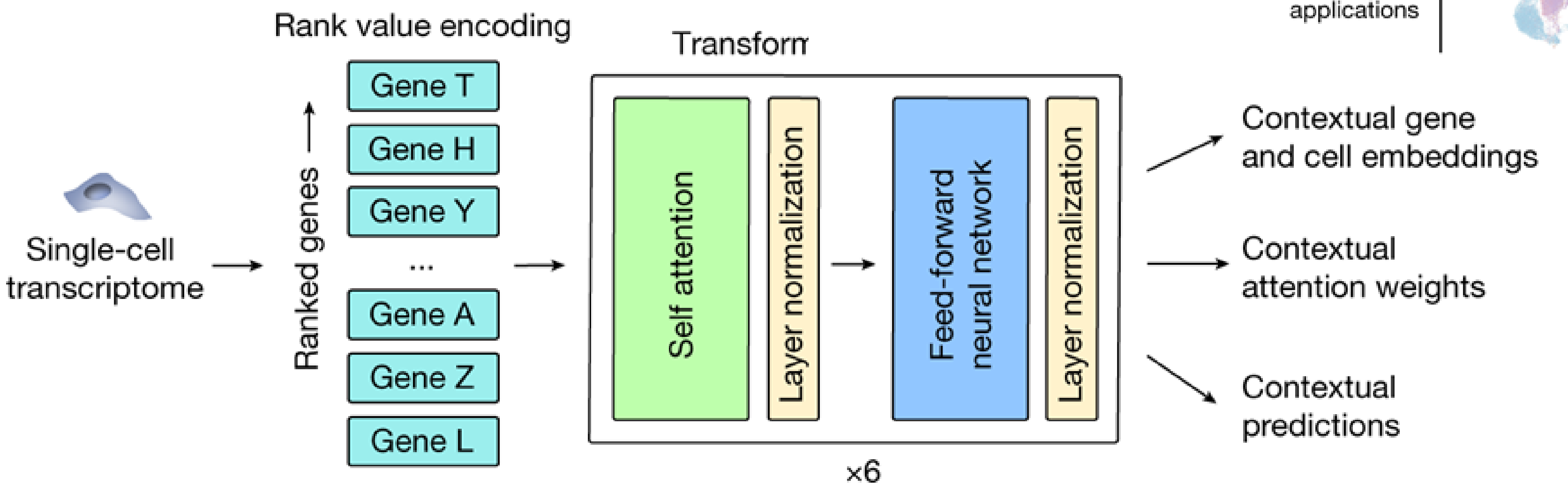
Train by PanglaoDB (***209 human single-cell datasets*** consisting of ***74 tissues***, and ***1,126,580 cells)***

Yang, F., Wang, W., Wang, F. et al. “scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data”. *Nat Mach Intell* (**2022**).

geneFormer

- Take raw count as input
- Embed expression on rank value only
- Average pool gene embedding to cell embedding
- Only published code at HuggingFace
- Vocabulary: 25424

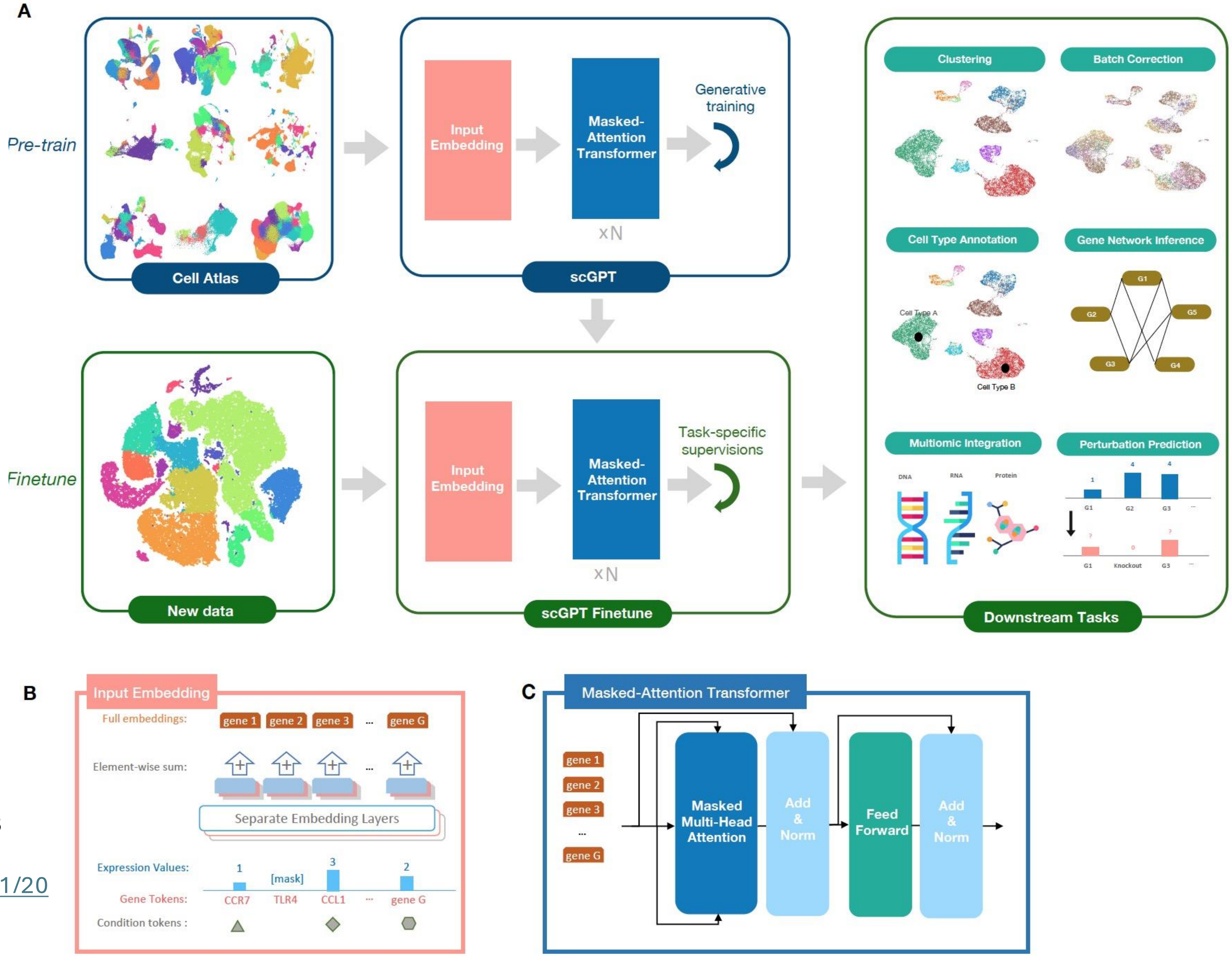




Theodoris CV, Xiao L, Chopra A, Chaffin MD, Al Sayed ZR, Hill MC, Mantineo H, Brydon EM, Zeng Z, Liu XS, Ellinor PT. Transfer learning enables predictions in network biology. Nature. 2023 May 31:1-9.

scGPT

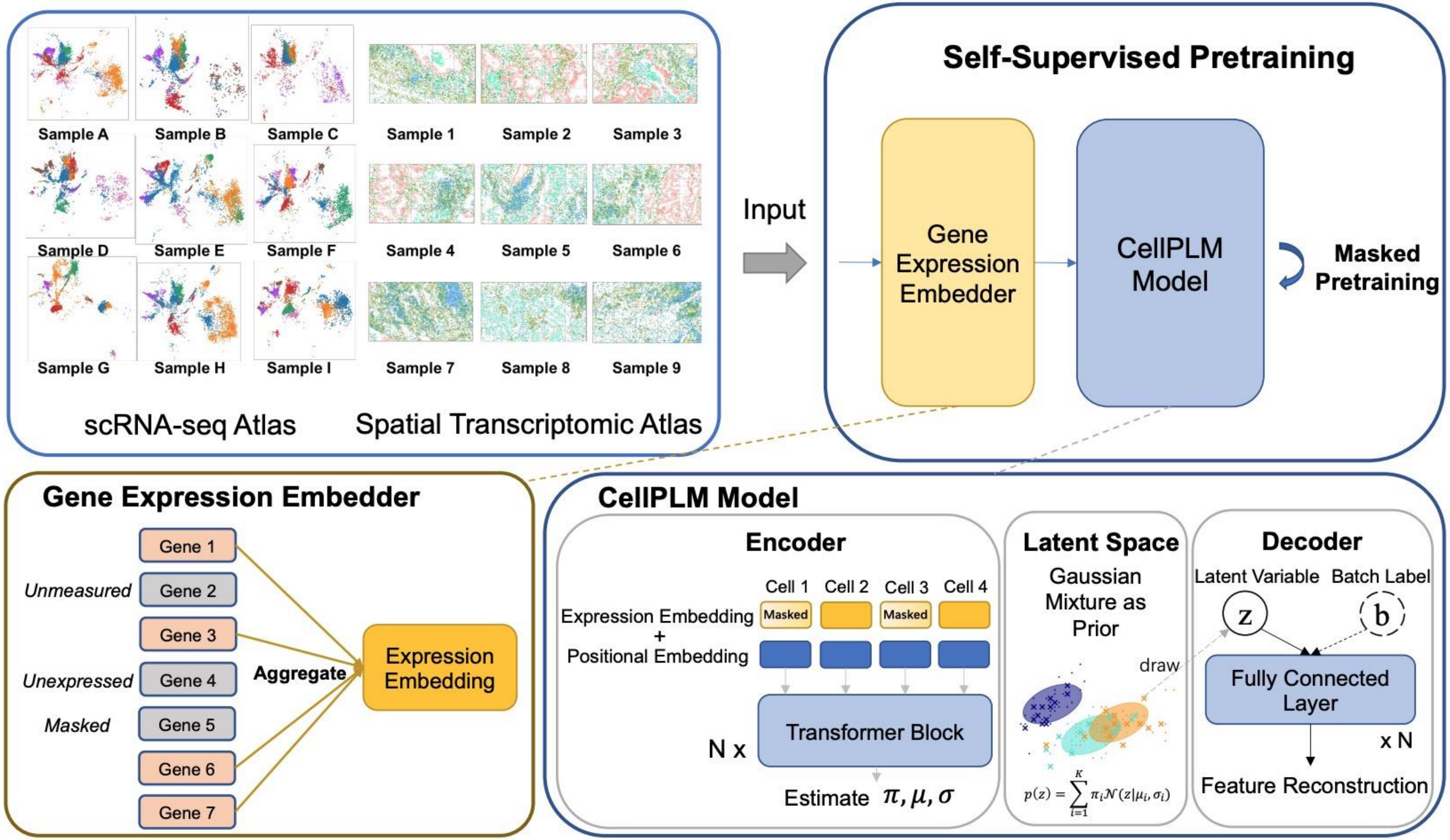
- Binning expression profile as well
- Set special token <CLS> token as cell embedding
- Fully open sourced and documented
- Vocabulary: 60697



Cui H, Wang C, Maan H, Pang K, Luo F, Wang B. scgpt: Towards building a foundation model for single-cell multi-omics using generative ai. bioRxiv. 2023 May
https://www.biorxiv.org/content/10.1101/2023.04.30.538439v2.full.pdf

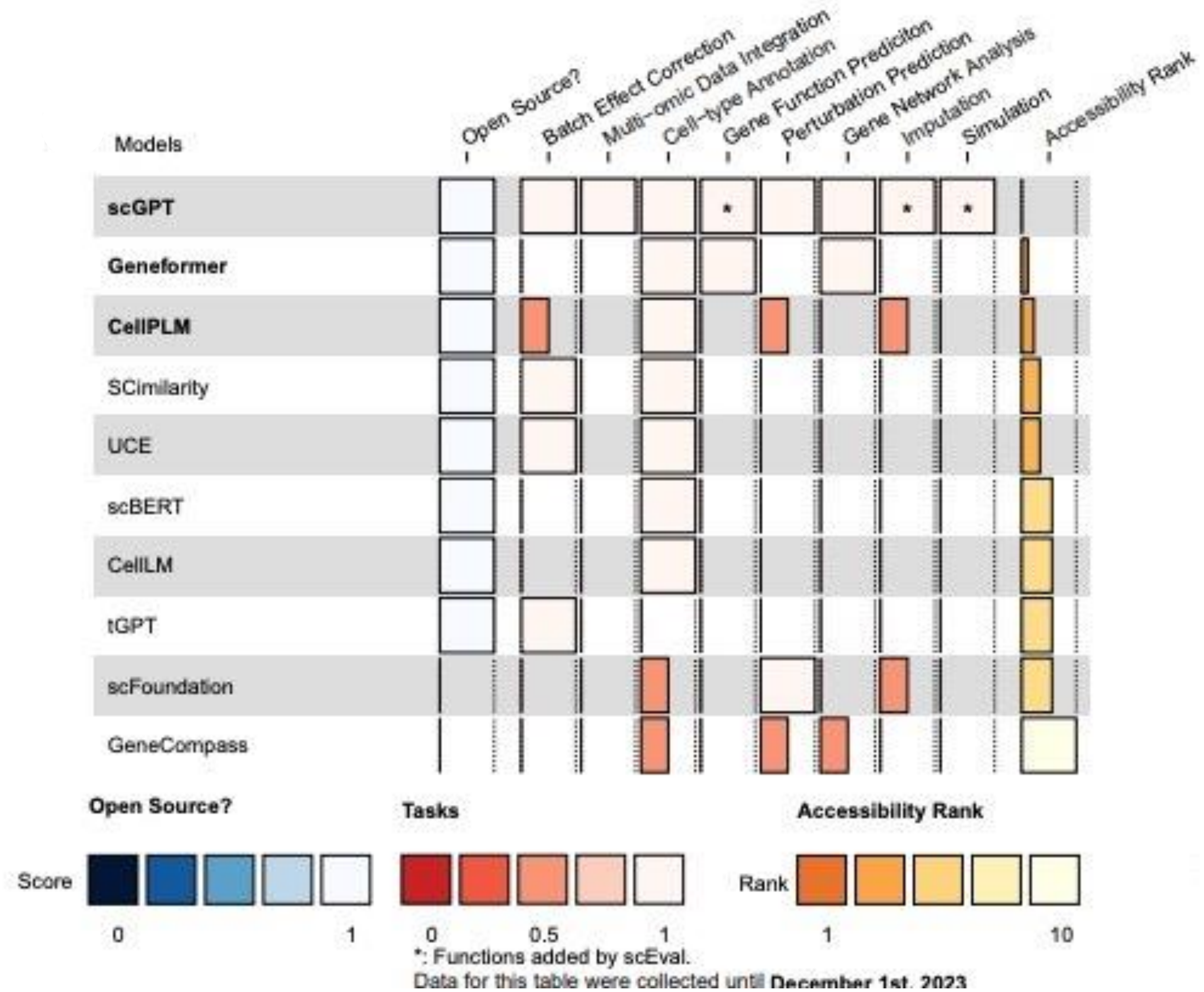
cellPLM

- Cell as token, tissue as sentence
- Unsupervised MLM (Masked Language Model) pre-training



Wen, Tang, Dai, Ding, Jin, Xie, Jiliang Tang. CellPLM: Pre-training of Cell Language Model Beyond Single Cells. bioRxiv 2023.

Overall evaluations of scLLMs from scEval



Tianyu Liu, Kexing Li, Yuge Wang, Hongyu Li , Hongyu Zhao. Evaluating the Utilities of Foundation Models in Single-cell Data Analysis. bioRxiv 2023.09.08.555192

Single-cell LLM Applications

Relatively mature

- cell type annotation
- reference mapping
- atlas retrieval
- cell-state embedding
- integration support
- few-shot transfer

Promising but mixed

- perturbation prediction
- drug response
- GRN inference
- cross-species transfer
- spatial niche prediction
- H&E-to-expression prediction

Still aspirational

- prompt-only analysis
- universal zero-shot reasoning
- causal mechanism discovery
- clinical-grade prediction
- full virtual cell simulation

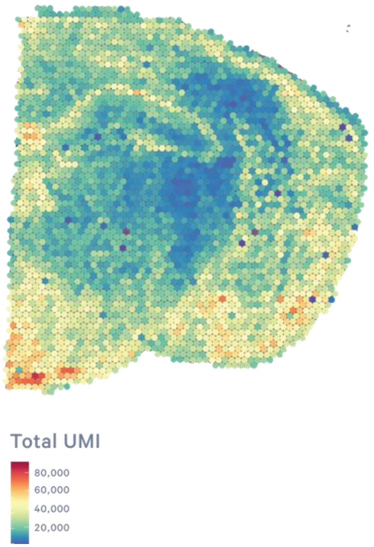
Freeze — use embeddings directly → **Probe** — linear model or small classifier → **Tune** — fine-tune all or selected layers → **PEFT** — LoRA / adapters / prompts → **Retrieve** — nearest cells + ontology + literature → **Continue** — pretrain on target tissue or platform

Design principle

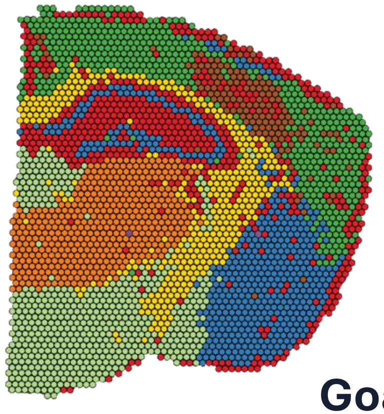
Use the cheapest adaptation that is adequate for the task.
Keep an external validation cohort.
Document model, data, and reference atlas versions.

Spatial Foundation Models

Visium whole transcriptome



Visium spot clusters



Input data
- gene expression
- x–y coordinates
- cell / spot graph
- optional H&E image

→

Pretrain at scale
- masked modeling
- contrastive learning
- multimodal alignment

→

Transfer tasks
- annotation
- domain discovery
- gene imputation
- image→gene prediction

Goal: learn reusable representations of cells, spots, niches and tissue regions so models transfer across organs, diseases and spatial platforms.

Cells / spots expression vector → **Neighborhood graph spatial adjacency** → **Context encoder Transformer / GNN** → **Embeddings cell • niche • region**

Token choices

gene-ranked “sentences” • binned genes • spots • cells • image patches

Spatial context

kNN / radius graphs • tissue regions • niche-level aggregation • SE(2)-aware geometry

Outputs

cell-type labels • domains • spatially variable genes • retrieval

Representative spatial transcriptomics foundation models

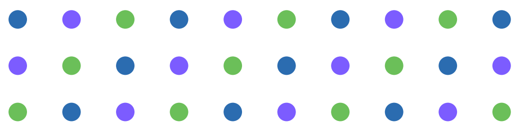
Model	Main modality	Core architecture	Typical outputs
Nicheformer	scRNA-seq + targeted spatial cells	Transformer trained on SpatialCorpus-110M	spatially aware cell embeddings; niche transfer
Novae	ST cell/spot graphs	Graph foundation model	zero-shot spatial domains; SVG/pathway analyses
OmiCLIP / Loki	H&E + Visium transcriptome	visual–omics contrastive model	alignment, annotation, deconvolution, image→gene
STPath	whole-slide images + ST	generative geometry-aware Transformer	zero-shot prediction of 38,984 genes
SToFM / HEIST	high-res ST graphs/multiscale data	SE(2) Transformer / hierarchical graph Transformer	region segmentation, annotation, imputation, clustering
scGPT-spatial / ST-Align	scGPT continual ST pretraining / image-gene pairs	Transformer alignment variants	cell typing, ST-specific embeddings, multimodal matching

Expression-and-neighborhood models: Nicheformer and Novae

Nicheformer

Transformer trained on both human/mouse single-cell and targeted spatial transcriptomics. Spatial Corpus >110M cells and 73 organs/tissues.

Best fit: spatially aware cell embeddings, transfer of niche information

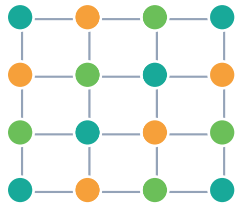


cell tokens with learned niche context

Novae

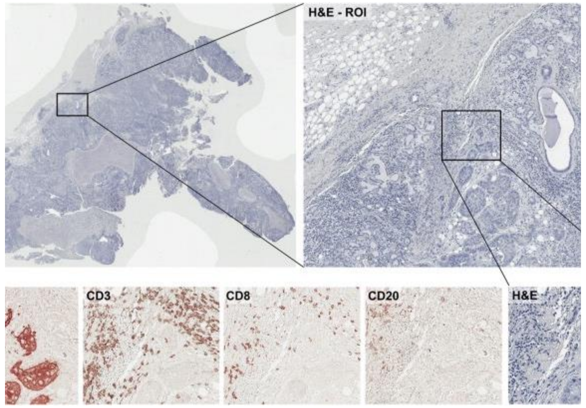
Graph-based foundation model for single-cell or spot-resolution ST. Designed for zero-shot spatial domain assignment across gene panels, tissues and technologies.

Best fit: spatial domain inference, batch correction, spatially variable genes/pathways, tissue architecture analysis.



spatial graph embeddings

H&E Foundation Models



Gigapixel whole slide imaging

tile, normalize, keep spatial coordinates

Self-supervised pretraining

DINO/MAE/contrastive learning at scale

Patch + slide embeddings

encode morphology, tissue context, report semantics

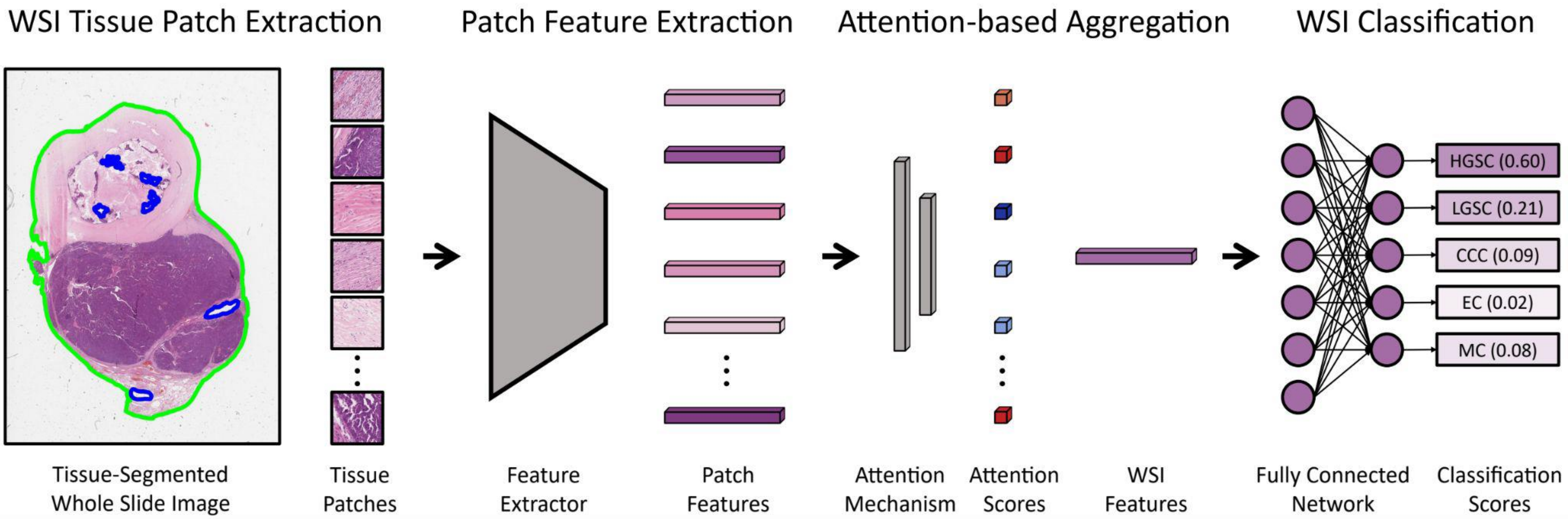
Adaptation

linear probe, fine-tune, retrieval, zero-shot

From morphology to embeddings

Large-scale pretraining turns routine H&E slides into general-purpose features for diagnosis, biomarker prediction, prognosis, retrieval, and multimodal reasoning.

H&E Foundation Models



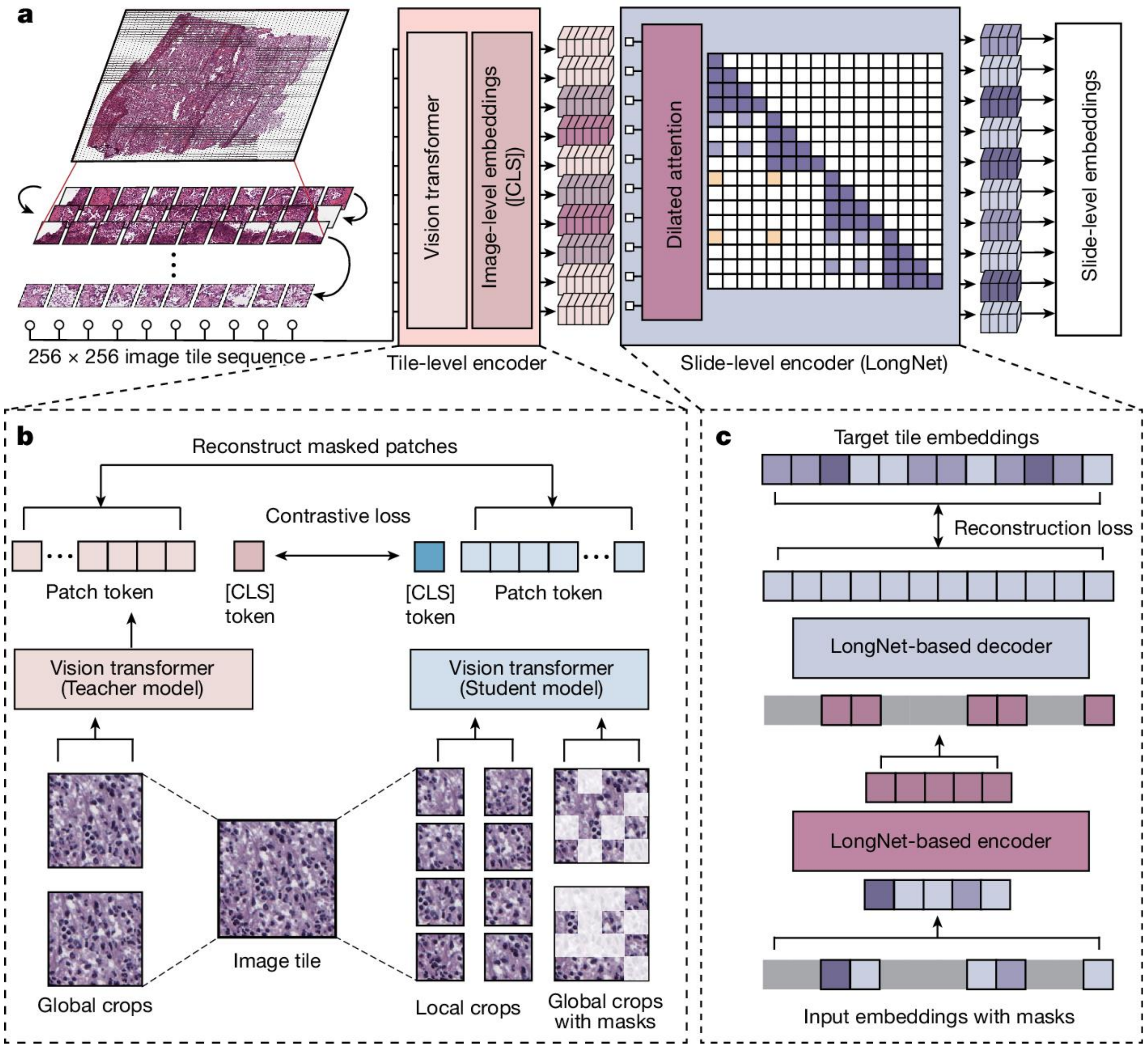
Jack Breen et al., npj Precision Oncology, 2025

Representative H&E / pathology foundation models

Model	Training signal	Scale / data	Best viewed as	Typical strengths
UNI / UNI2	Self-supervised vision	>100M H&E patches from >100k WSIs; UNI2 >200M H&E/IHC images	General-purpose patch encoder	Broad tissue generalization; strong feature extractor
CONCH	Vision-language contrastive	1.17M histopathology image–caption pairs	Patch-level V-L model	Retrieval, zero-shot, captioning, segmentation/classification
Virchow / Virchow2	DINOv2-style SSL	Virchow2: 3.1M WSIs; 632M / 1.85B params	Scaled patch encoder	High-performing morphology/biomarker features
Prov-GigaPath	Tile + whole-slide pretraining	1.3B tiles from 171,189 WSIs, 31 tissue types	Whole-slide encoder	Slide-level tasks; ultra-large-context modeling
H-Optimus-0	SSL vision	1.1B params; >500k H&E slides	Large open histology encoder	Strong open-source baseline at scale
TITAN / PRISM	Multimodal slide-report alignment	TITAN: 335,645 WSIs + reports + synthetic captions; PRISM uses Virchow embeddings + reports	Slide-level V-L models	Text prompting, report generation, zero/few-shot slide reasoning

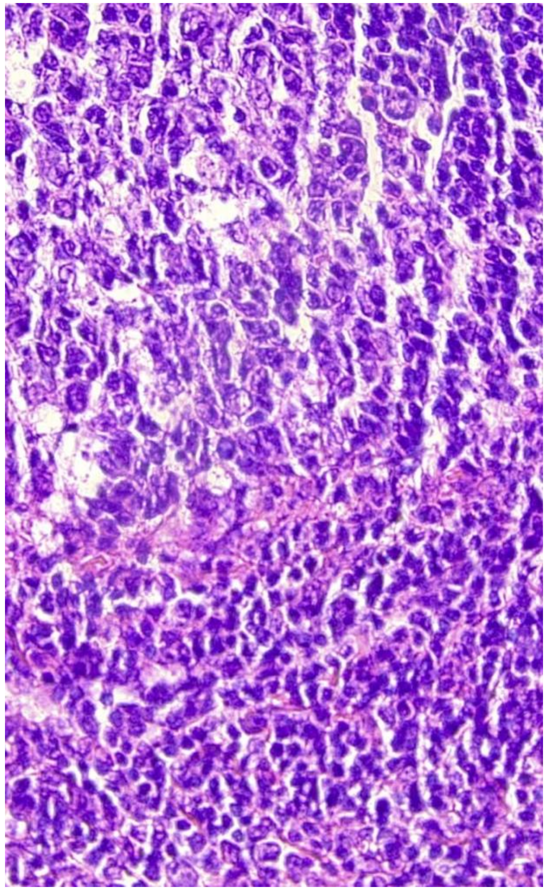
Prov-GigaPath

- Tile-encoder trained with DINOv2
- Slide-encoder trained with masked autoencoder
- Combines local tile representations with slide-level aggregation using a transformer architecture designed for gigapixel pathology.
- Trained on 171k WSIs from 30k patients



Source: Xu, A whole-slide foundation model for digital pathology from real-world data, 2024 (Microsoft)

Multimodal whole-slide models



TITAN

Transformer-based pathology Image and Text Alignment Network.

Pretrained on 335,645 WSIs using visual self-supervision plus report/caption alignment; uses 423,122 synthetic captions.

PRISM

Slide-level multimodal generative model built on Virchow tile embeddings and clinical report text.

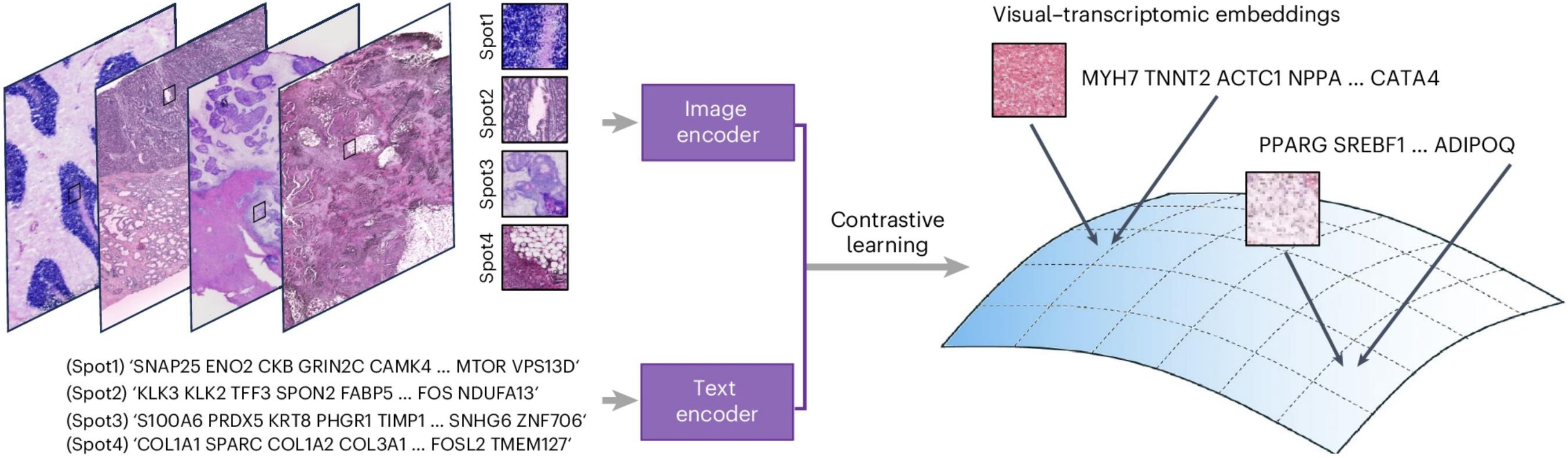
Supports slide embeddings, text prompting, zero-shot cancer detection/subtyping, and report-style generation.

Why it matters

Diagnostic categories, uncertainty, rare entities, and biomarkers are often encoded in language.

Multimodal FMs can connect morphology with clinical semantics.

OmiCLIP



•Visual–omics foundation model that integrates H&E with spatial transcriptomic profiles.
•Transcriptomic profiles were converted into gene "sentences" by concatenating the top-expressed genes within each spatial patch.
•trained on a large-scale dataset of 2.2 million paired image–transcriptomics patches across 32 organs.
•Built on OmiCLIP, the Loki platform supports tissue alignment, tissue annotation, cell-type decomposition, image–transcriptomics retrieval, and gene expression prediction from H&E images.

Perturbation-aware modeling: virtual cells

Input perturbation space

gene knockout / CRISPRi
drug exposure
dose and time
cell type and state
microenvironment

Modeling options

FM embedding + predictor
graph neural models
causal / mechanistic priors
retrieval from perturbation atlases
uncertainty estimates

Validation needs

held-out perturbations
held-out cell types
time-course validation
external datasets
experimental confirmation

A model or model stack that represents a cell state, accepts interventions or environmental changes, and predicts downstream molecular or phenotypic responses.

- encode a cell state
- simulate perturbation or context changes
- generate uncertainty-aware predictions
- support hypothesis generation and prioritization

From cell embeddings to virtual-cell systems

Curated multi-omics atlases cells, spatial sections, perturbations, pathology, clinical metadata

Foundation representations gene, cell, niche, image and patient embeddings

Biological priors pathways, GRNs, protein interactions, ontologies, literature

Adaptation + reasoning fine-tuning, retrieval, causal models, AI agents

Validated outputs hypotheses, biomarkers, perturbation responses, trial stratification

Virtual-cell pipeline for single-cell biology

Reference atlases
healthy, disease, developmental and treatment states

Trajectory / time-course data
state transitions, differentiation, activation, recovery

Perturb-seq and drug-response data
CRISPR, knockout, CRISPRi, compounds, dose and time

Spatial and microenvironment context
neighbors, niches, pathology image context, tissue regions

Multi-omics and prior knowledge
ATAC, proteins, pathways, GRNs, PPIs, literature

State encoder
single-cell / spatial / multi-omics foundation representations

Latent cell state
cell identity, state, trajectory, niche, uncertainty

Perturbation engine
edit genes, drugs, dose, time, environment, combinations

Response decoder
predict expression, pathways, fate, phenotype, sensitivity

Validation loop
held-out perturbations, time points, cell types, wet-lab follow-up

Potential outputs

- counterfactual gene expression
- predicted response trajectories
- responder / non-responder hypotheses
- candidate regulators or combinations

Translational opportunities

Disease subtyping

cell-state-specific patient stratification

Rare cells

detect rare but clinically meaningful states

Immunotherapy

tumor microenvironment and immune activation

Autoimmunity

progression and preclinical signatures

Drug response

screening, repurposing and toxicity hypotheses

Pathology bridge

connect H&E morphology to molecular programs

Key limitations

- out-of-distribution perturbations
- sparse perturbation coverage
- mechanism vs correlation
- compute and evaluation burden

Arc Institute virtual-cell model: State

Single-cell gene expression

State centers on single-cell RNA-seq because gene expression is a scalable, largely unbiased readout of cell state. The basic task is: given a starting transcriptome and a perturbation, predict the resulting change in RNA expression.

Training data scale

- ~170 million observational cells
- >100 million perturbational cells
- 70 cell lines
- includes stem, cancer, and immune cells
- leverages large perturbation atlases such as Tahoe-100M

Model view of transcriptomes

State uses two linked modules: a State Embedding (SE) module that maps transcriptomes into a smooth latent space, and a State Transition (ST) module that predicts how cells move through that space after genetic, chemical, or cytokine perturbation.

Gene-expression-centric workflow

Input — single-cell transcriptome + perturbation

Learn — cell-state manifold from massive data

Predict — post-perturbation RNA-expression shift

Use — drug discovery and hypothesis prioritization

Summary

- scFMs are powerful reusable representation engines, not magic biological reasoners.
- Model families differ: gene-token, atlas-query, biology-informed, perturbation, spatial/multimodal, virtual cell.
- Better in signal-to-noise ratios, holistic features, batch effects…
- Representation choice defines what biology the model can see.
- Adaptation and validation are usually the path to useful performance.
- The field is moving toward spatially grounded, perturbation-aware virtual-cell systems.